

Q. Given,

```
int fib(int n){  
    if (n == 0)  
        return 0;  
    else if (n == 1)  
        return 1;  
    else  
        return fib(n-1) + fib(n-2);  
}
```

Let,

$n = \$t_0$ address ~~pointers~~

~~return~~ return value = \$58

return value = \$t₄, pointer = \$p

Mips code:

Fun:

	1
addi \$p, \$p, -8	2
sw \$a0, 4(\$p)	3
sw \$t0, 0(\$p)	4
beq \$t0, \$0, r1	5
li \$t1, 1	6
beq \$t0, \$t1, r2	7
sub \$t0, \$t0, \$t1;	8
Jal Fun;	9
add more \$t3, \$t4, \$0	10
lw \$t0, 0(\$p)	11
li \$t1, 2	12
sub \$t0, \$t0, \$t1	13

o jal	fun	14
add	\$t4, \$t4, \$t3	15
jr	end	16
<u>r1:</u>		17
li	\$t4, 0	18
jr	end	19
		20
<u>r2:</u>		
li	\$t4, 2	21
jr	end	22
<u>end:</u>		23
lw	\$a8, 0(\$p)	24
addi	\$p, \$p, 8	25
jr	\$a8;	26

—X—

total time = 26